



Time-Scale Rescaling in Early Universe Structure Growth

Álvaro Quiceno

Abstract

This short note isolates a single claim inside RTM and pushes it into a cosmology-adjacent, falsifiable back-of-the-envelope. If characteristic process times scale as $T \propto L^\alpha$, then in an early universe with much smaller environmental scale L_{env} , effective times shorten. Taking L_{env} to track the Hubble scale L_H in a minimal "FRW+ α " ansatz yields a simple acceleration factor A by which any mesoscopic timescale is divided. Evaluated at $z \sim 10$ this gives order-of-magnitude speed-ups of **20-37 $\times$** for $\alpha \sim 1$, consistent in direction with "too-early/too-massive" galaxies. We then show, parametrically, how large a speed-up A would be needed to reproduce stellar masses/luminosities like those reported at $z > 10$ without touching BBN/CMB: the trick is to keep α inactive (band near 0) in the homogeneous plasma era and active (order-unity) only in multiphase, structured baryonic media.

Preliminary empirical validation $\rightarrow$ (APPENDIX B). We validate the time-rescaling hypothesis using a catalog of 55 stellar mass estimates from galaxies observed by JWST (JADES, CEERS, Labbé et al. 2023, UNCOVER, and GLASS) at redshifts $z = 6.0$ to 16.4 . Initial heuristic analysis finds that 44% of cataloged galaxies exceed standard Λ CDM limits, yielding an apparent coherence exponent of $\alpha = 1.33 \pm 0.30$. After injecting continuous ± 0.3 dex SED fitting variance and Eddington bias correction via Monte Carlo simulation, the bias-corrected exponent converges to $\alpha = 1.16 \pm 0.08$, statistically distinguishable from $\alpha = 1.0$ ($p < 10^{-6}$). The excess- z trend — correlation between redshift and the gap between observed and Λ CDM-predicted stellar mass — yields Spearman $\rho = 0.43$, $p = 0.006$ (calibration-independent). These results are consistent with the early universe operating in a high-coherence topological regime ($\alpha > 1$). **Flanking campaign (May 2026) $\rightarrow$ (APPENDIX D):** three robustness tests on the 55-galaxy catalog confirm the excess- z signal is not a cross-survey artifact (significant in 3/4 surveys independently), that the excess grows with redshift at fixed stellar mass (Spearman $\rho = +0.761$, $p < 10^{-6}$, mass-controlled), and that $A(z, \alpha = 1)$ is sufficient to resolve all 55 galaxies without requiring $\alpha > 1$.

1) Minimal ansatz: FRW+ α with $L = H^{-1}$

$$T \propto L^\alpha$$

Choose the **environmental scale** L to be the FRW Hubble length $H^{-1}(z)$. Define the **operational time rescaling** between a small interval of standard cosmic time dt and the process' "effective" time $d\tau$:

$$d\tau = \left(\frac{L(z)}{L_0} \right)^\alpha dt = \left(\frac{H_0}{H(z)} \right)^\alpha dt$$

Equivalently, any process timescale $\tau_{std}(z)$ (computed in standard physics) is **accelerated** by

$$\tau_{RTM}(z) = \frac{\tau_{std}(z)}{A(z; \alpha)}, \quad A(z; \alpha) \equiv \left(\frac{H(z)}{H_0} \right)^\alpha$$

where $A(z; \alpha)$ is the **RTM acceleration factor**.

With Λ CDM background,

$$\frac{H(z)}{H_0} = [\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda]^{1/2}$$

At $z \gtrsim 10$ (matter-dominated to good approximation),

$$A(z; \alpha) \simeq \sqrt{\Omega_m} (1+z)^{3/2}$$

$$\frac{H(z)}{H_0} \simeq \sqrt{\Omega_m} (1+z)^{3/2} \Rightarrow$$

2) Worked numbers at $z=10$: why "20–40×" often appears

Two reference choices:

Einstein–de Sitter toy ($\Omega_m=1$)

$$A_{\text{EdS}} = (1+z)^{\alpha/2}$$

At $z=10$ and $\alpha=1$:

$$A_{\text{EdS}} = (1+10)^{3/2} = 11^{1.5} \approx \mathbf{36.5}$$

Hence $A \approx 37$: processes **$\sim 37\times$ faster** than today (at the same class/scale).

"Realistic" Λ CDM ($\Omega_m=0.315$, $\Omega_\Lambda=0.685$)

$$A_{\Lambda\text{CDM}} = [\Omega_m(1+z)^3 + \Omega_\Lambda]^{\alpha/2}$$

For $z=10$, $\alpha=1$:

$$A_{\Lambda\text{CDM}} = [0.315 \times 11^3 + 0.685]^{1/2} \approx \mathbf{20.5}$$

For $z=7$, $\alpha=1$:

$$A_{\Lambda\text{CDM}} \approx \mathbf{12.7}$$

Interpretation: the factor "37×" is the pedagogical EdS limit; in the current Λ CDM the number is $A \sim 20$ for $z \sim 10$ with $\alpha \sim 1$. In either case, the order of magnitude **$A \sim 20\text{--}40$** emerges immediately.

3) Galaxy assembly: required acceleration A (closed formula)

Consider a halo with mass M_h and baryon fraction $f_b \approx 0.157$

Let ε_{dyn} be the efficiency per dynamical time (fraction of gas converted into stars per t_{dyn}) and N the number of dynamical times available between the onset of the cold phase and the redshift of interest:

$$N \equiv \frac{\Delta t(z)}{t_{\text{dyn},\text{std}}(z)}$$

If the per-step conversion is independent (minimal model), the integrated efficiency after N steps is:

$$SFE_{\text{std}} = 1 - (1 - \varepsilon_{\text{dyn}})^N \approx 1 - e^{-\varepsilon_{\text{dyn}}N} \quad (\varepsilon_{\text{dyn}} \ll 1)$$

The expected stellar mass is:

$$M_*^{\text{std}} \approx f_b M_h SFE_{\text{std}}$$

Under RTM, the effective number of steps grows by the factor A :

$$N_{\text{RTM}} = AN, \quad \Rightarrow \quad SFE_{\text{RTM}} = 1 - (1 - \varepsilon_{\text{dyn}})^{AN} \approx 1 - e^{-\varepsilon_{\text{dyn}}AN}$$

To reach a target stellar mass M_*^{tgt} at redshift z :

$$A_{\text{req}} \geq \frac{1}{\varepsilon_{\text{dyn}}N} \ln \left[\frac{1}{1 - \frac{M_*^{\text{tgt}}}{f_b M_h}} \right];$$

$$N = \frac{\Delta t(z)}{t_{\text{dyn},\text{std}}(z)}$$

3.1) Back-of-the-envelope numbers (illustrative)

- $z = 14$: cosmic age $\Delta t \sim 0.28 - 0.30$ Gyr.
- Halo dynamical time: $t_{\text{dyn},\text{std}} \sim \kappa H^{-1}(z)$ with $\kappa \approx 0.1$ (virial density $\sim 200\rho_m$)

In Λ CDM:

$$H(z)/H_0 \approx 31.8 \Rightarrow t_{\text{dyn}} \approx 0.1/31.8 H_0^{-1} \approx 44 \text{ Myr.}$$

$$\Rightarrow N \approx \Delta t/t_{\text{dyn}} \approx 300/44 \approx 6.8.$$

Case A (demanding):

$$M_h = 10^{11} M_\odot \Rightarrow f_b M_h = 1.57 \times 10^{10} M_\odot$$

$$\text{Target } M_*^{\text{std}} = 10^{10} M_\odot \Rightarrow SFE_{\text{req}} \approx 0.637$$

If $\varepsilon_{\text{dyn}} = 0.01$ (1% per t_{dyn}):

$$A_{\text{req}} \gtrsim \frac{1}{0.01 \times 6.8} \ln \left(\frac{1}{1 - 0.637} \right) \approx 14.7 \times 1.01 \approx 15$$

$\Rightarrow$ With $\alpha = 1$:

- **EdS**: $A \approx 37$ (ample margin)
- **Λ CDM**: $A \approx 32$ (also sufficient)

With $A \sim 37$ (EdS) or $A \sim 20$ (Λ CDM), the required acceleration $A_{req} \sim 10-15$ is still achievable with margin. For the most demanding cases ($M_{star} \sim 10^{11}$ at $z > 12$), $\alpha \sim 1.2$ may be needed in Λ CDM.

Case B (moderate): same configuration but $\varepsilon_{dyn} = 0.02$:

$$A_{req} \approx \frac{1}{0.136} \times 1.01 \approx 7.5$$

Here $\alpha \sim 0.5$ could already suffice ($A \approx 5.6 - 7.67$, depending on the background).

Moral: with efficiencies per t_{dyn} in the range 1 – 2% and massive halos ($10^{10} M_{\odot}$), an acceleration $A \sim 7 - 15$ makes $M_* \sim 10^{10} M_{\odot}$ at $z \sim 14$ arithmetically plausible **without** touching the FRW background or “breaking” anything; α in 0.7 – 1.0 delivers it naturally.

4) Does it break BBN/CMB? No, if α obeys “complexity bands”

To avoid altering nucleosynthesis and recombination:

- **Band hypothesis (RTM):** $\alpha \approx 0$ for homogeneous plasma (BBN/CMB era, low morphological complexity); $\alpha \sim O(1)$ only emerges in multiphase baryonic media (cold gas + turbulence + cooling + feedback), i.e., *after* the dawn of structure.
- **EFT companion:** choose portals and ξ (non-minimal $\alpha^2 R$) within the safe wedge so that α does not modify early expansion or atomic physics beyond EP/PPN/BBN/CMB limits.

This allows α to act as a **mesoscopic time rescaling factor** (cooling, collapse, feedback cycles), **not** as exotic background energy.

5) Predictions and tests (how to falsify the hypothesis)

1. Time-scale relation within the same z : at $z \approx 10 - 15$, processes with effective spatial scale L (e.g., star-forming regions) should show:

$$T(L) \propto L^{\alpha}, \text{ with } \alpha \approx 0.7 - 1.0 \text{ if the case requires } A \gtrsim 10$$

Observationally: durations of bursts, outflow escape times, etc., as a function of size.

2. **Apparent efficiencies:** for the same M_h , the integrated efficiency SFE should be higher at high z due to the effective A factor (equation for A_{req}). If A is small, high SFE is not reached without fine-tuning.
3. **No touching BBN/CMB/PPN:** no α effect should appear in background linear observables; all the novelty should occur at mesoscopic scales post-collapse. (This is testable in the EFT companion with the “safe wedge”.)

6) Limitations (what we do not solve here)

- We do not derive $\alpha(z)$ from microphysics nor solve FRW with backreaction of α ; we use $L = H^{-1}$ as an environmental proxy.

- We do not compute the luminosity function or SED spectra; we only show the time kinematics and a bound on the required acceleration.
- The "37×" number is the EdS limit; the realistic value for Λ CDM is **A~20** at $z\sim 10$ with $\alpha\sim 1$.

7) Executive summary

With $L_{\text{env}} = L_{\text{H}}$ and $\alpha\sim 1$, the acceleration factor is

$$A = (H(z)/H_0)^\alpha$$

At $z=10$: - $\alpha=1 \Rightarrow \mathbf{A\approx 37}$ (EdS) or **A~20** (Λ CDM) - $\alpha=1.5 \Rightarrow A\approx 220$ (EdS) or $A\approx 91$ (Λ CDM)

The required acceleration to reach target M_{star} is

$$A_{\text{required}} = \ln[1 - M_{\text{star}}/(f_{\text{b}} \cdot M_{\text{halo}})] / [N_{\text{dyn}} \cdot \ln(1-\epsilon)]$$

With $M_{\text{halo}}\sim 10^{12} M_{\odot}$, $\epsilon\sim 2\%$, and $N_{\text{dyn}}\sim 5$, **A~10–20 suffices** for $M_{\text{star}}\sim 10^{11} M_{\odot}$.

This is compatible with $\alpha\sim 1$ without touching BBN/CMB, if α is off in homogeneous plasma and on only in complex media (RTM bands).

Appendix A

Table 1: RTM Acceleration Factor $A(z)$ for $\alpha=1$

Redshift z	Cosmic Age (Λ CDM)	A_{EdS}	$A_{\Lambda\text{CDM}}$
5	1.17 Gyr	14.7	8.3
7	0.76 Gyr	22.6	12.7
10	0.47 Gyr	36.5	20.5
12	0.37 Gyr	46.9	26.3
15	0.27 Gyr	64.0	35.9
20	0.18 Gyr	96.2	54.0

EdS: $A = (1+z)^{(3/2)}$. Λ CDM: $A = [0.315(1+z)^3 + 0.685]^{(1/2)}$. Planck 2018 parameters.

Appendix B: JWST Empirical Validation of Time-Scale Rescaling

The recent deployment of the James Webb Space Telescope (JWST) has revealed a population of unexpectedly massive galaxies at high redshifts ($z > 10$). Under the standard Λ CDM cosmological model, assuming a linear progression of cosmic time, these structures appear too massive to have formed within the available temporal window, creating a profound tension in modern astrophysics. The Rhythmic Multiscale Transport (RTM) framework provides a natural resolution: at high redshifts, the universe existed in a more "coherent" topological state ($\alpha > 1$), accelerating the dynamics of structure formation.

B.1 Heuristic Analysis (Point-Estimate Observation)

We compiled a catalog of 55 high-redshift galaxies from recent JWST surveys (JADES, CEERS, UNCOVER, GLASS). By defining an "Acceleration Factor" required to reconcile the observed stellar masses with the theoretical specific star formation rate limits, we extracted the implied coherence exponent (α) for each galaxy.

The initial point-estimate analysis demonstrates that 44% of the cataloged galaxies (24 out of 55) strictly exceed standard Λ CDM limits. Averaging these direct observations yields an apparent exponent of $\alpha = 1.33 \pm 0.30$ ($p < 0.0001$). While visually compelling, relying solely on point-estimates in high-redshift astrophysics can be susceptible to observational artifacts, necessitating a more rigorous statistical treatment.

B.2 Rigorous Probabilistic Validation (Monte Carlo & Bias Correction)

To ensure the RTM signal is a genuine physical law and not a statistical illusion caused by measurement noise, we subjected the catalog to a rigorous probabilistic stress-test. Two major astrophysical confounding variables were introduced into the model:

1. **SED Fitting Variance:** Typical stellar mass estimates at $z > 10$ carry massive uncertainties. We injected a continuous ± 0.3 dex variance into all mass readings.
2. **Eddington / Selection Bias:** The tendency for surveys to preferentially detect overluminous (and seemingly overmassive) outliers at the edge of instrumental sensitivity.

We deployed a Monte Carlo simulation generating 10,000 parallel universes, mathematically smoothing the mass distributions to absorb these observational biases.

B.3 Conclusion of the JWST Anomaly

After bias correction, the standard Λ CDM assumption of purely linear time ($\alpha = 1.0$) is statistically disfavored ($p < 10^{-6}$ against the bias-corrected exponent $\alpha = 1.16 \pm 0.08$). The Monte Carlo distribution converges stably on this bias-corrected value.

The excess- z correlation — Spearman $\rho = 0.43$, $p = 0.006$ — is the most calibration-independent result in this analysis. It measures whether the gap between observed and Λ CDM-predicted stellar mass increases with redshift, which is the directional prediction of RTM time-rescaling. This correlation survives SED uncertainty injection and is classified as **NOVEL** by the Red Team (April 2026): it is not predictable from Λ CDM alone.

Interpretation caveat: the $p < 10^{-6}$ value reflects statistical separation of $\alpha = 1.16$ from $\alpha = 1.0$ within the Monte Carlo model. It does not account for the possibility that the input stellar masses carry systematic errors beyond the 0.3 dex injection (e.g., non-parametric SED biases at $z > 12$, photometric redshift catastrophic failures, or AGN-boosted luminosities). These systematic uncertainties are the primary limitation of the analysis and are the target of the spectroscopic falsification tests in Section 5. Full audit results and limitations: Appendix C.

APPENDIX C — Red Team Audit: Verification and Certification (April 2026)

The empirical claims in this document were subjected to independent adversarial audit by the RTM Red Team using **Claude Opus 4.6 with Extended Thinking** in April 2026. The following verification record is provided for transparency.

C.1 What Was Tested

Claim	Test	Result
44% of galaxies exceed Λ CDM limits	Direct count from catalog	Confirmed ✓
Apparent $\alpha = 1.33 \pm 0.30$ (heuristic)	Point-estimate computation	Confirmed ✓
Bias-corrected $\alpha = 1.16 \pm 0.08$	Monte Carlo 10,000 iterations	Confirmed ✓
$p < 10^{-6}$ vs. $\alpha = 1.0$	Statistical test within MC model	Confirmed within model ✓
Excess-z trend $\rho = 0.43$, $p = 0.006$	Spearman correlation across 55 galaxies	Confirmed — most robust result ✓
BBN/CMB safety (band hypothesis)	Theoretical consistency check	Internally consistent ✓
$A \sim 20\text{-}37\times$ at $z = 10$	Analytical derivation from FRW+ α	Confirmed ✓

C.2 Classification Verdict

Finding	Classification	Rationale
Excess-z correlation ($\rho = 0.43$, $p = 0.006$)	NOVEL	Not predictable from Λ CDM; directional RTM prediction confirmed
Bias-corrected $\alpha = 1.16 \pm 0.08$	NOVEL (exploratory)	Model-dependent on SED uncertainty assumption
$A \sim 20\text{-}37\times$ acceleration factor at $z = 10$	CONSISTENT	Order-of-magnitude agreement with observational tension
BBN/CMB safety via complexity bands	FALSIFIABLE	Testable via EFT companion (Section 5)
44% of galaxies exceed Λ CDM	CONVERGENT	Consistent with Labbé et al. 2023, Boylan-Kolchin 2023

C.3 Key Limitation Identified

The Red Team identified one structural limitation that is not present in the original document:

The $p < 10^{-6}$ value is computed **within the Monte Carlo model** — it measures the probability of observing $\alpha \geq 1.16$ if the true value were $\alpha = 1.0$, given the ± 0.3 dex noise injection. However:

1. The 0.3 dex SED uncertainty is a representative central estimate. Some JWST photometric redshift measurements at $z > 12$ carry uncertainties of 0.5 – 1.0 dex (Steinhardt et al. 2023, Adams et al. 2023). At larger injected variance, the statistical significance would decrease.
2. The Eddington bias correction assumes the selection function is well-characterized. For JWST early science, this may not be fully established.
3. AGN contamination at high- z is not modeled. If a fraction of “massive galaxies” are AGN-dominated, the excess is partly observational rather than astrophysical.

None of these limitations invalidate the excess- z trend ($\rho = 0.43$, $p = 0.006$), which is calibration-independent. They affect the precision of the bias-corrected exponent ($\alpha = 1.16 \pm 0.08$).

C.4 Tone Corrections Applied

Original phrase	Corrected to
“categorically rejected”	“statistically disfavored”
“conclusively validates the RTM prediction”	“consistent with the RTM prediction”
“definitively rule out Eddington bias”	“correct for Eddington bias within the model”
“firmly rejects”	“is statistically distinguishable from”
“This confirms with high statistical significance”	“These results are consistent with”

C.5 Surviving Novel Finding

The most defensible RTM contribution in this document is the **excess- z correlation (Spearman $\rho = 0.43$, $p = 0.006$)**. This is:

- Calibration-independent (does not require assuming a specific SED uncertainty level)
- Directionally predicted by RTM time-rescaling (more excess at higher z)
- Not predictable from standard Λ CDM without additional physics
- Falsifiable by the spectroscopic tests in Section 5

The bias-corrected exponent $\alpha = 1.16 \pm 0.08$ is the model-dependent expression of the same signal. Both survive the Red Team audit. The primary uncertainty is whether future spectroscopic confirmation reduces the apparent excess (weakening the finding) or confirms it (strengthening it).

C.6 Red Team Verdict

Score: 70% — PASS. The core findings (excess- z $\rho = 0.43$, bias-corrected $\alpha = 1.16 \pm 0.08$) are statistically sound within their model assumptions. The acceleration factor derivation ($A \sim 20\text{-}37\times$ at $z = 10$) is analytically correct. The BBN/CMB safety argument via complexity bands is internally consistent. The document correctly identifies Section 5 falsification tests — spectroscopic mass confirmation, process timescale measurements, and absence of α effects in background observables — as the next required step.

The primary limitation (systematic uncertainty beyond the 0.3 dex injection) is acknowledged and does not invalidate the finding, but it prevents the stronger interpretation (“categorical rejection of Λ CDM”) from being supported. The exploratory, falsifiable framing of this document is its greatest strength.

Appendix D: Flanking Campaign — Robustness Validation (May 2026)

This appendix presents three robustness tests applied to the 55-galaxy JWST catalog. All computations are reproducible via `rtm_jwst_flanks.py` with the accompanying `jwst_galaxy_catalog.csv`.

D.1 Subsample Robustness by Survey (Flank A)

The 55 galaxies originate from four independent surveys (JADES, CEERS, UNCOVER, GLASS). If the excess- z correlation is a cross-survey artifact from combining heterogeneous datasets, it should disappear within individual surveys.

Survey	n	Excess galaxies	ρ	p	Significant?
CEERS	15	6/15	+0.654	0.008	✓
JADES	8	0/8	+0.762	0.028	✓
Other	25	13/25	+0.441	0.027	✓
UNCOVER	5	2/5	+0.600	0.285	ns (n=5)

3 of 4 testable surveys show significant positive excess- z correlation. The signal survives within individual surveys — it is not a cross-survey selection artifact. UNCOVER (n=5) is underpowered.

D.2 Excess Grows with Redshift at Fixed Stellar Mass (Flank B)

A simple excess- z split by redshift bin shows a selection effect: high- z galaxies in this catalog have lower stellar masses (observational bias toward the most massive objects at each redshift). After controlling for stellar mass via linear residualization:

$$\text{excess}_{\text{residual},i} = \text{excess}_i - (\hat{c}_0 \cdot \log M_{*,i} + \hat{c}_1)$$

Spearman $\rho(z, \text{excess}_{\text{residual}}) = +0.761, p < 10^{-6}$.

At fixed stellar mass, higher-redshift galaxies deviate MORE from the Λ CDM prediction. This is the directional prediction of RTM time-rescaling: the acceleration factor $A(z) = [H(z)/H_0]^\alpha$ grows with z , so the Λ CDM deficit should increase at higher z for a given mass. The mass-controlled correlation confirms this.

Bin	n	Mean excess (raw)	$\rho(z, \text{excess})$	Note
Low- z ($z \leq 9.0$)	28	+0.086	+0.533	Weak
High- z ($z > 9.0$)	27	+0.541	+0.054	Weak (selection effect)
Mass-controlled	55	—	+0.761	Strong ($p < 10^{-6}$)

The raw high- z correlation is weak due to the observational selection effect. After mass control, the signal is the strongest in the entire analysis.

D.3 Acceleration Factor Sufficiency (Flank C)

For each galaxy, we compute the RTM acceleration factor available at its redshift ($A_{\Lambda\text{CDM}} = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}^\alpha$ with $\alpha = 1$) and compare it to the required acceleration to reconcile the galaxy with ΛCDM . RTM predicts: galaxies requiring the most acceleration should cluster where $A(z)$ is largest (high z).

Classification	n	Mean z	Mean excess (dex)	Mean A_{avail}
Impossible (excess > 0.5)	13	10.7	+1.15	22.9
Tension ($0 < \text{excess} \leq 0.5$)	27	10.1	+0.23	21.1
Consistent (excess ≤ 0)	15	7.3	−0.29	13.4

Key finding: $A(z, \alpha = 1)$ is **sufficient for 55/55 galaxies** — no galaxy requires $\alpha > 1$ to be resolved. “Impossible” galaxies cluster at $z \approx 10.7$ (highest A_{avail}), consistent with RTM. “Consistent” galaxies cluster at $z \approx 7.3$ (lower A_{avail}).

D.4 Summary

Flank	Result	Key metric	For RTM
D.1 By survey	POSITIVE	3/4 surveys significant	Not a cross-survey artifact
D.2 Mass-controlled excess- z	POSITIVE	$\rho = +0.761, p < 10^{-6}$	Excess grows with z as predicted
D.3 Acceleration sufficient	POSITIVE	55/55 resolved with $\alpha = 1$	No exotic $\alpha > 1$ needed

All three flanks support the RTM interpretation. The mass-controlled excess- z correlation ($\rho = +0.761$) is the strongest result in this document — it shows that the JWST anomaly is not simply “massive galaxies exist at high z ” but that “at fixed mass, higher- z galaxies deviate more from ΛCDM ,” the directional prediction of RTM time-rescaling.